ASTR-8750: Machine Learning for Physical Sciences (Fall 2025)

Location: Kinard 116 (K-116)

Time: Monday 3:00 – 4:00 PM

Credits: 1

Instructor: Srinadh Reddy Bhavanam, Postdoctoral Fellow, Marco Ajello's High-Energy Astrophysics Group. *Research area: Applied ML for Astrophysics*

Email: srinadb@clemson.edu

Office hours: K-120A, Friday, 10:00 – 11:00 AM

Prerequisites

This course requires students to have knowledge of linear algebra (vectors, matrices, eigenvalues/eigenvectors), calculus (derivatives, gradients, chain rule), probability & statistics (random variables, distribution, Bayes' theorem) and Python programming.

Course Description

Graduate course on practical machine learning for physics and astronomy. Covers an end-to-end workflow (from problem framing to evaluation), with hands-on supervised/unsupervised methods and a brief neural-networks intro, emphasis on scientific interpretability, and rigorous evaluation metrics.

1. **Supervised Learning** (classification & regression)
    - Algorithms: k-Nearest Neighbours, Linear/Logistic Regression, Decision Trees, Random Forests, Support Vector Machines
    - Metrics: Accuracy, Precision/Recall, F1, ROC–AUC, RMSE/MAE

2. **Unsupervised Learning**
    - Clustering: k-means, DBSCAN
    - Dimensionality reduction: PCA
    - Anomaly/rare-event detection
3. **Learning Theory**
    - Bias–variance tradeoff, under/overfitting
    - Feature engineering and data preprocessing
4. **Deep Learning**
    - Perceptron and Multi-Layer Perceptron (MLP)
    - Optimization basics (SGD/Adam), initialization, activation functions
    - Training practice: early stopping, regularization, monitoring

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Learning Goals

By the end of the course, you will be able to:

- Understand ML at a practical level
- Able to build and evaluate ML solutions
- Ready to start ML research in your area of interest.

Evaluation

Final Project (100%): Choose a physics/astronomy dataset and complete an end-to-end ML analysis.

Deliverables:

- **Jupyter Notebook (.ipynb):** Executable, with data loading, preprocessing, model training/validation, and figures.
- **Project Report (PDF):** 6–10 pages including background, data description, methods, results (with appropriate metrics), error/uncertainty discussion, and scientific interpretation.

Milestones & Dates: announced in class.